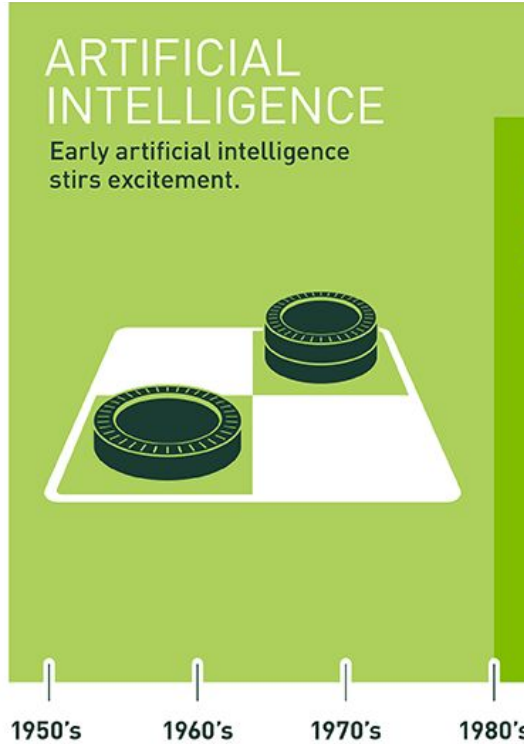
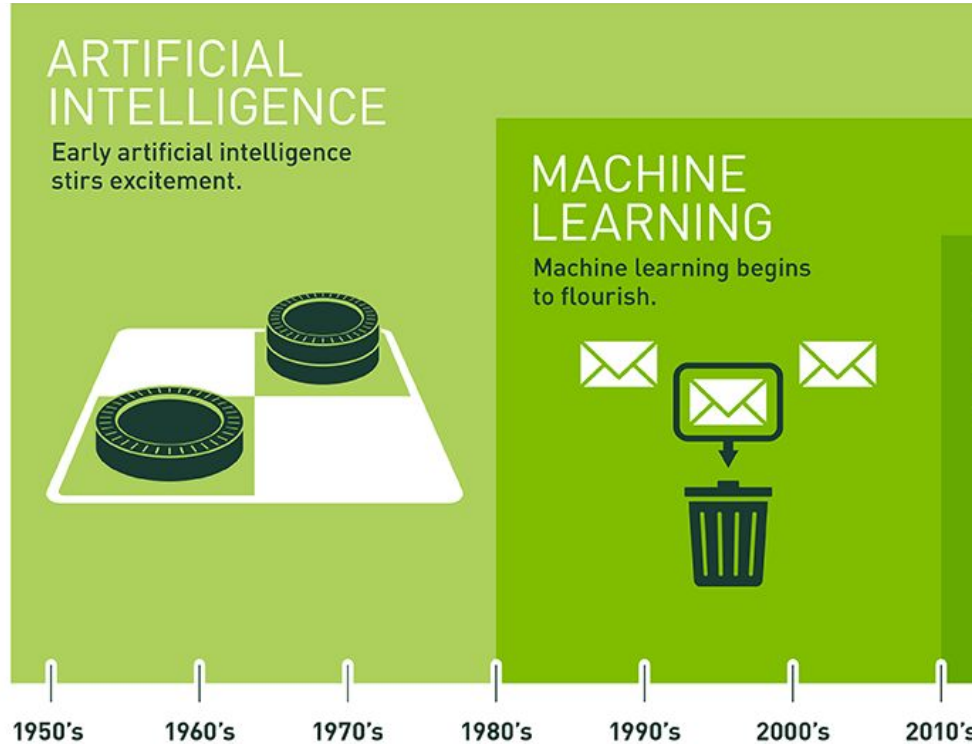


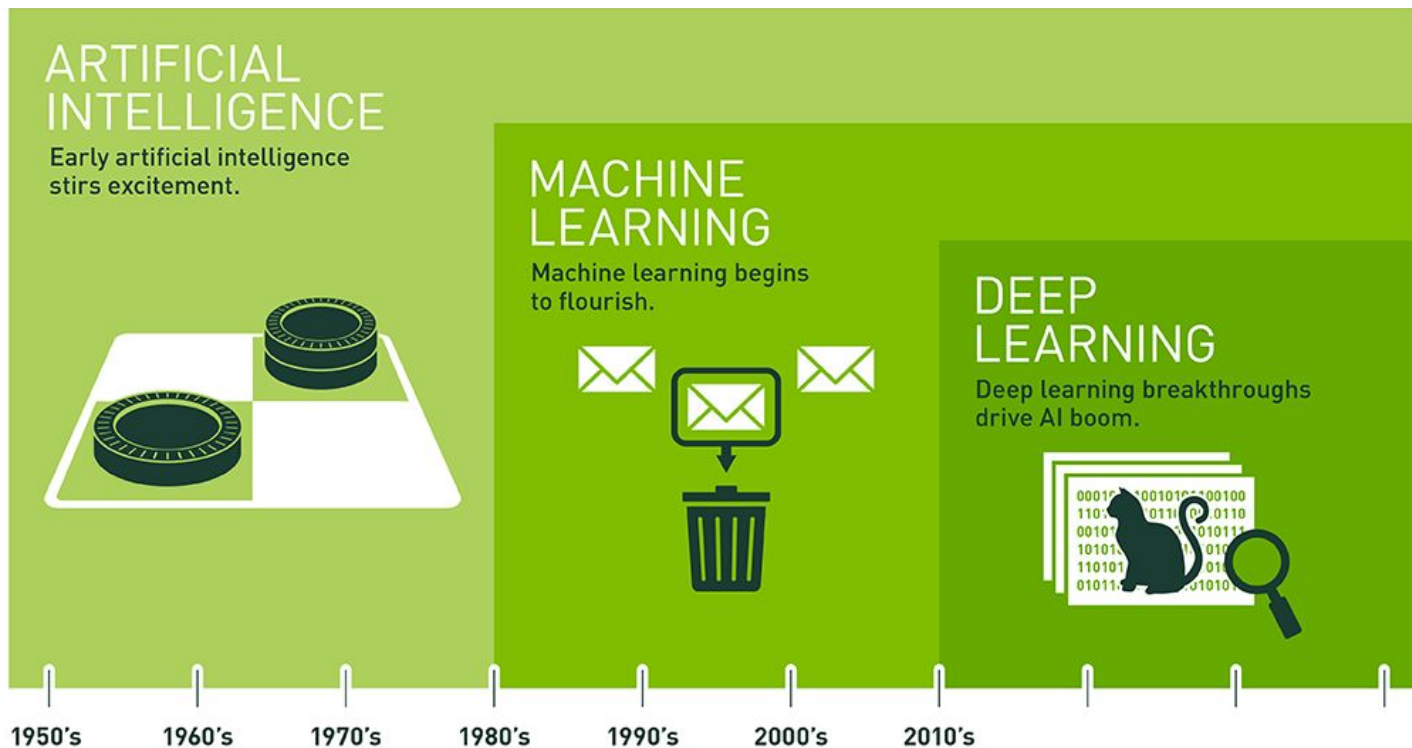
What is AI?



What is AI?



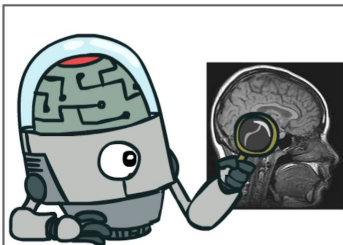
What is AI?



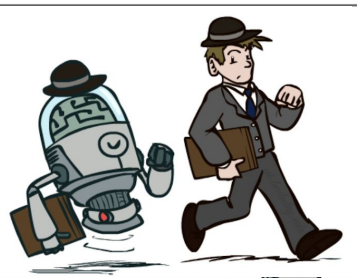
Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

What is AI?

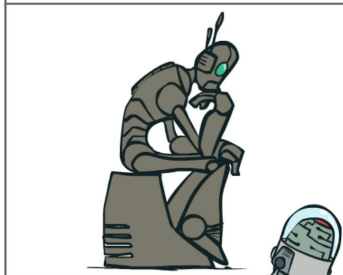
Thinking
humanly



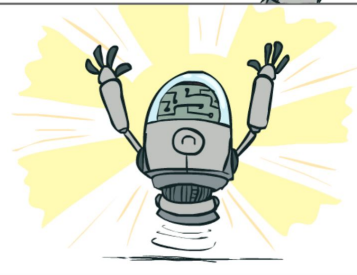
Acting
humanly



Thinking
rationally



Acting
rationally



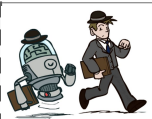
<http://www-inst.eecs.berkeley.edu/~cs188/fa19/>
<http://slazebni.cs.illinois.edu/fall17/>

What is AI?

Thinking
humanly

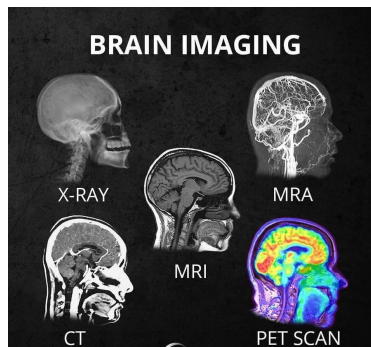


Acting
humanly



Make computers think like a human
How does a human think?

- brain imaging - neuroscience
- psychological experiments - cognitive science



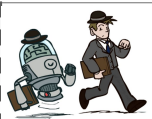
Alison Gopnik - developmental scientist

What is AI?

Thinking
humanly



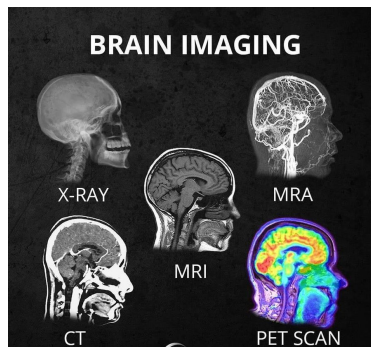
Acting
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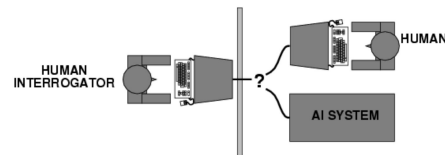
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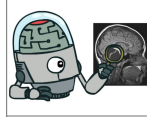
Alison Gopnik - developmental scientist

The Turing Test Approach



What is AI?

Thinking
humanly

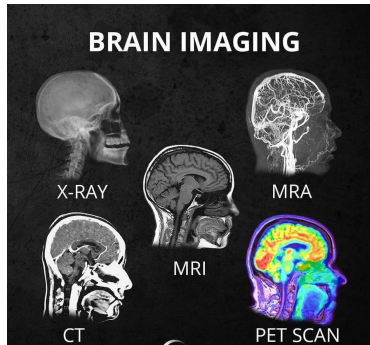


Acting
humanly



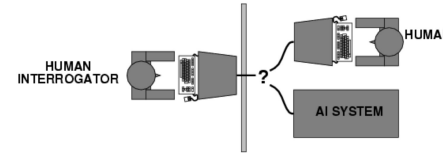
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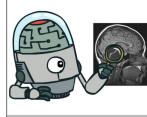
The Turing Test Approach



- **natural language processing** - communicate
- **knowledge representation** - store knowledge
- **automated reasoning** - answer questions and draw conclusions
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- **robotics** - manipulate objects

What is AI?

Thinking
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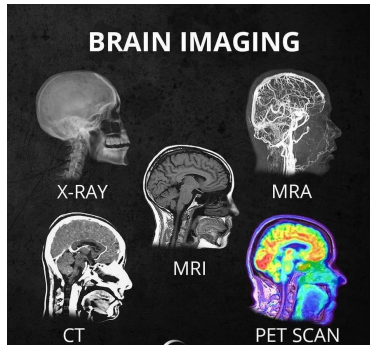


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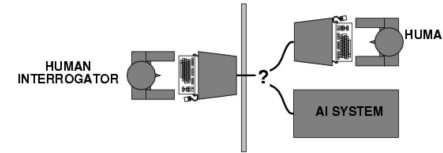
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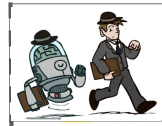
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Aeronautical engineering *"goal of the field is making machines that flies so exactly like pigeons that they can fool other pigeons"*

What is AI?



Acting
humanly

Issues with this test:

- Variability in protocols, judges
- Success depends on deception!
- Chatbots can do well using “cheap tricks” (Eliza 1966)

Winograd schema: Multiple choice questions that can be easily answered by people but cannot be answered by computers using “cheap tricks”

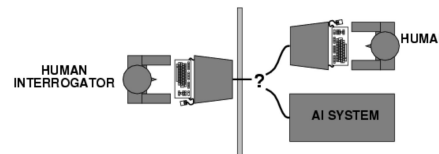
The sack of potatoes had been placed below the bag of flour, so it had to be moved first. What had to be moved first?

- The sack of potatoes
- The bag of flour

<https://ptchallenge-workshop.github.io/>



The Turing Test Approach

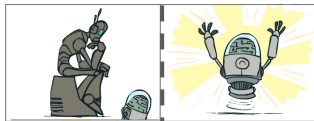


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Aeronautical engineering “*goal of the field is making machines that flies so exactly like pigeons that they can fool other pigeons*”

What is AI?

Thinking
rationally



Acting
rationally

Idealized or “right” way of thinking

Logic: patterns of argument that always yield correct conclusions when supplied with correct premises

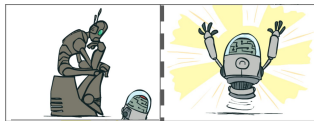
Logicist approach to AI: describe problem in formal logical notation and apply general deduction procedures to solve it

Issues:

- Describing real-world problems and knowledge in logical notation
- Dealing with uncertainty
- Computational complexity of finding the solution starting from a list of facts

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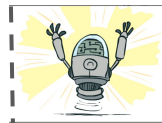
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A rational agent acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

- Goals are expressed in terms of the utility of outcomes (quantitative measure)
- Naturally accommodates uncertainty
- Rationality only concerns what decisions are made (no cognitive process behind)
- Being rational means maximizing your expected utility

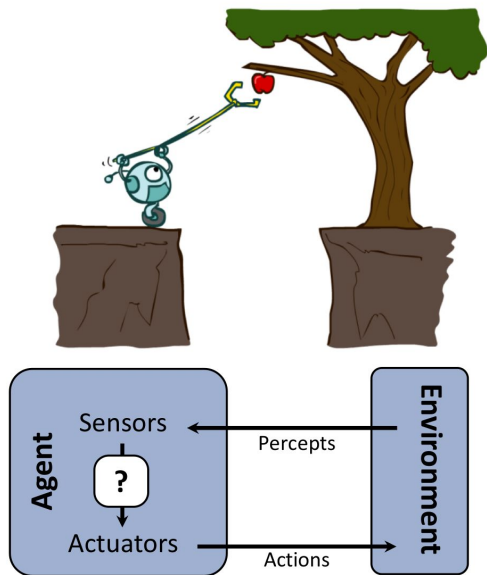
In practice, utility optimization is subject to the agent’s computational constraints (bounded rationality or bounded optimality)

What is AI?



Acting
rationally

AI = Computational Rationality



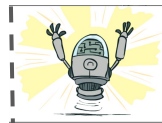
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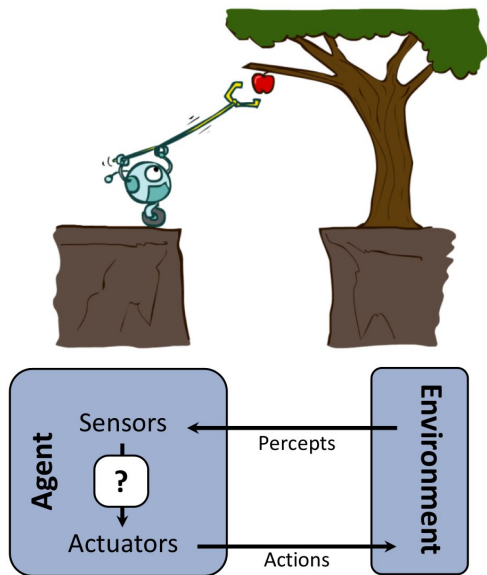
<http://www-inst.eecs.berkeley.edu/~cs188/fa19/>

What is AI?



Acting
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AI = Computational Rationality



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All the skills needed for the Turing Test allow an agent to act rationally

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- **automated reasoning** - answer questions and draw conclusions

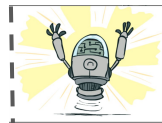
enable the agent to reach good decisions

- **natural language processing** - communicate
generate comprehensible sentences and get by in a complex society

- **machine learning** - detect and extract patterns, adapt to new circumstances
improve ability to generate effective behavior

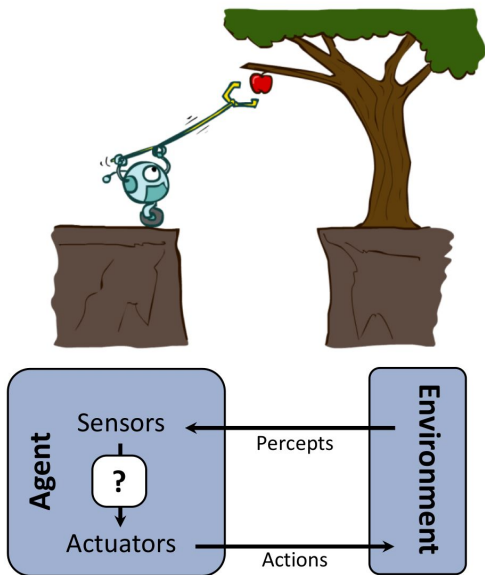
- **computer vision** - perceive objects
- **robotics** - manipulate objects
interact beyond language communication

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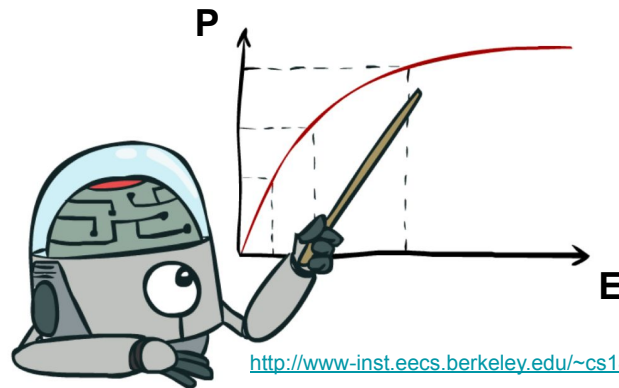
Machine Learning

How do we define “Learning”?

1. experience E (data)
2. task T
3. performance measure P

An agent learns when its performance at task T , measure by P , improves with experience E .

We need to get a computer do well on a task without manually building its competence.



Machine Learning

How do we define “Learning”?

1. experience E (data)
2. task T
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T : Playing checkers

E : Playing practice games against itself

P : Percentage of games won against an arbitrary opponent

T : Recognizing hand-written words

E : Database of human-labeled images of handwritten words

P : Percentage of words correctly classified

T : Driving on four-lane highways using vision sensors

E : A sequence of images and steering commands recorded while observing a human driver.

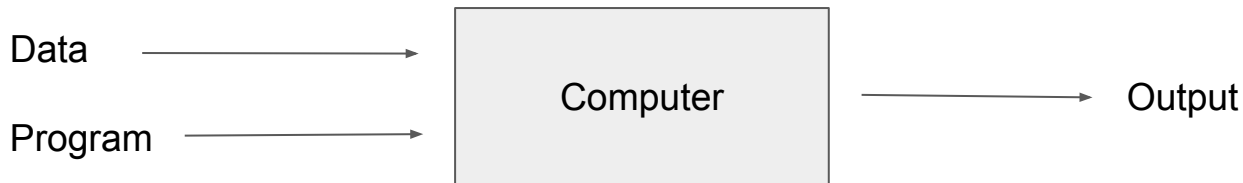
P : Average distance traveled before a human-judged error

T : Categorize email messages as spam or legitimate

E : Database of emails, some with human-given labels

P : Percentage of email messages correctly classified

Programming with Data

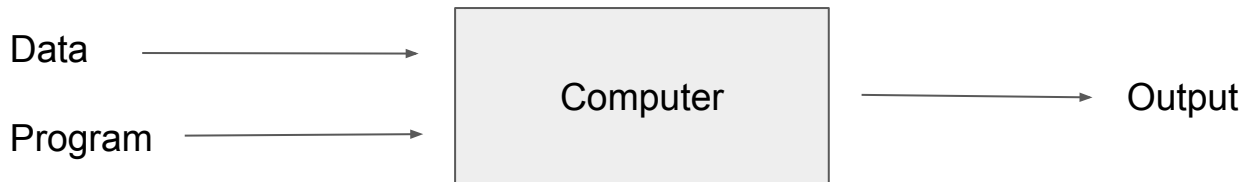


We want adaptive, robust and fault tolerant systems

Rule based implementation (often)

- is difficult (for programmers)
- is brittle (can miss many edge cases)
- becomes a nightmare to maintain explicitly
- does not work too well

Programming with Data



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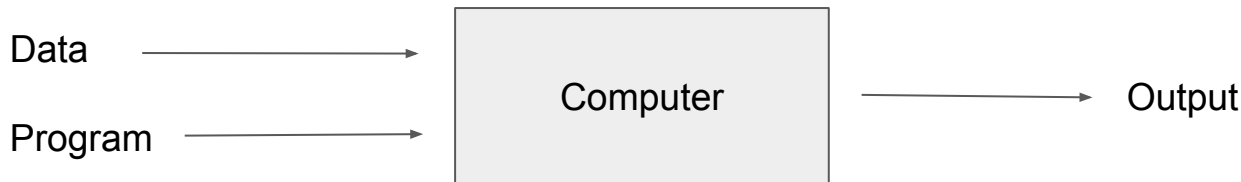
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figure credit: G. Hinton

Programming with Data



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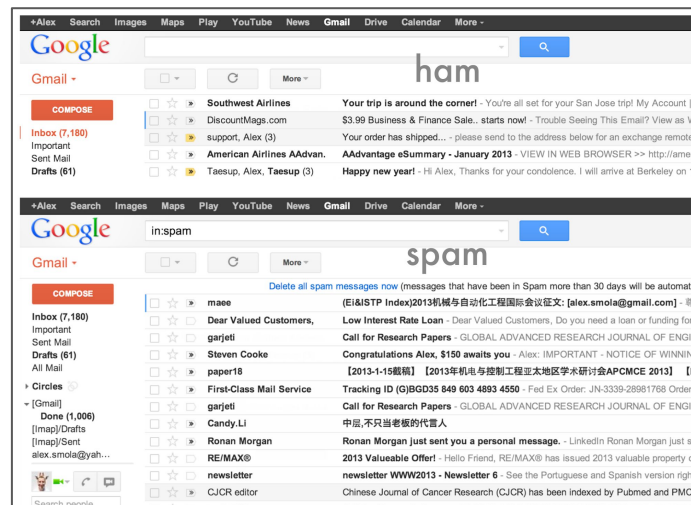
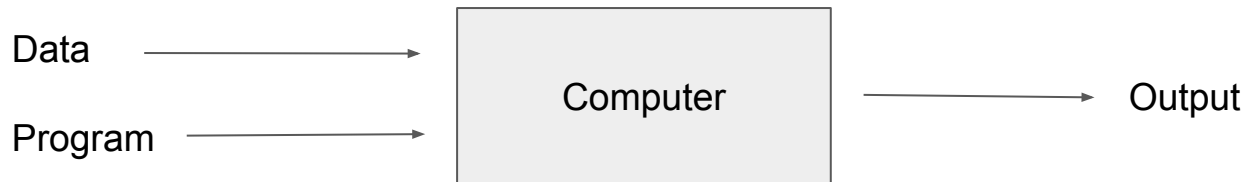


figure credit: A. Smola

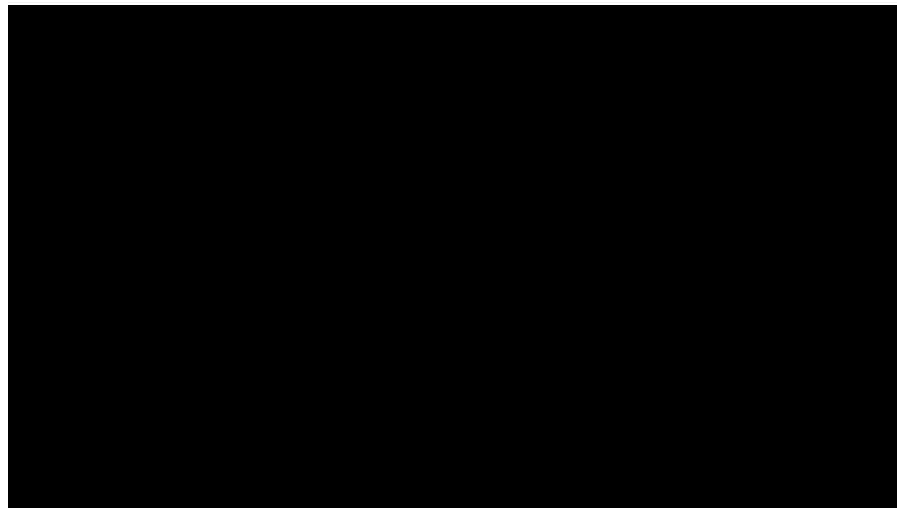
Programming with Data



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NETFLIX

Netflix Prize

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NETFLIX

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Movies For You

Randy, the following movies were chosen based on your interest in:

- [Bowling for Columbine](#)
- [Carnivale: Season 1](#)
- [Fahrenheit 9/11](#)



The Big One

★★★★★
A riveting and subversive look at the events of 9/11 from the perspective of the people who were there.



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You really liked it...

Now only for just \$5.99

[Shop](#) as low as \$5.99

The Netflix Prize seeks to substantially improve the accuracy of predictions about how much someone is going to love a movie based on their movie preferences. Improve it enough and you win one (or more) Prizes. Winning the Netflix Prize improves our ability to connect people to the movies they love.

Read the [Rules](#) to see what is required to win the Prizes. If you are interested in joining the quest, you should [register a team](#).

You should also read the [frequently-asked questions](#) about the Prize. And check out how various teams are doing on the [Leaderboard](#).

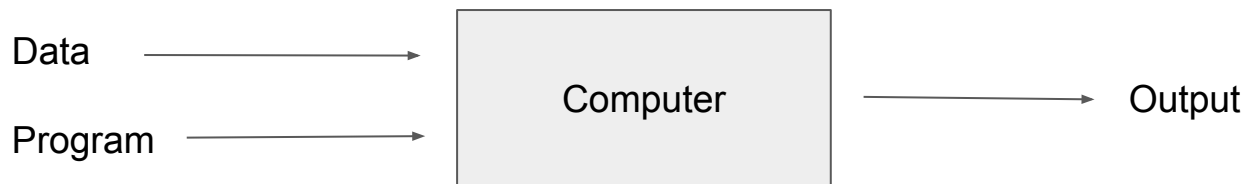
Good luck and thanks for helping!

[FAQ](#) | [Forum](#) | [Netflix Home](#)

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Programming with Data

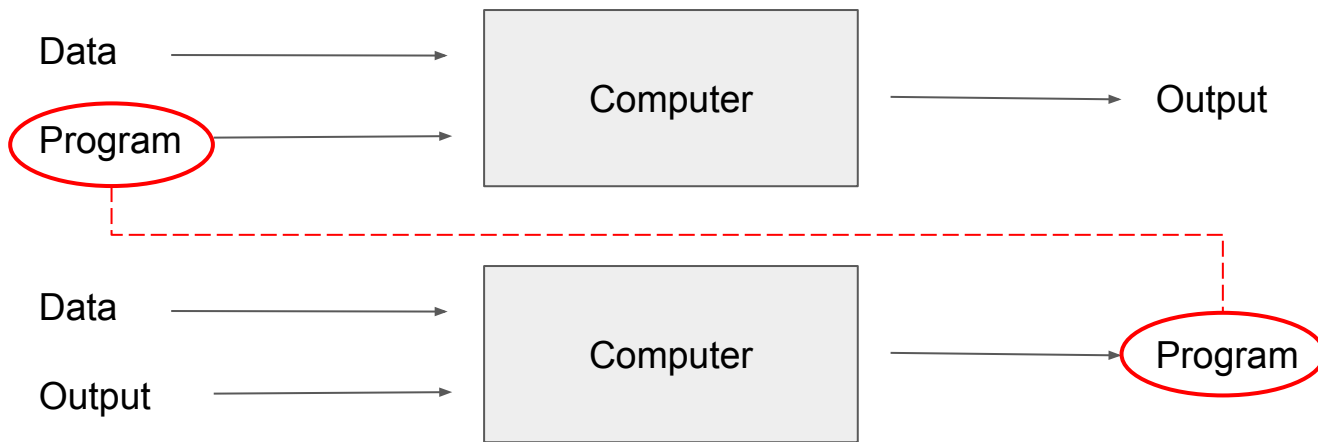


We want adaptive, robust and fault tolerant systems

Rule based implementation (often)

- is difficult (for programmers)
 - is brittle (can miss many edge cases)
 - becomes a nightmare to maintain explicitly
 - does not work too well
-
- It is easier to obtain examples of what we want: **IF x THEN DO y**
 - collect many pairs (x_i, y_i)
 - estimate a function f such that $f(x_i) = y_i$ (supervised learning)
 - detect patterns in data (unsupervised learning)

Programming with Data

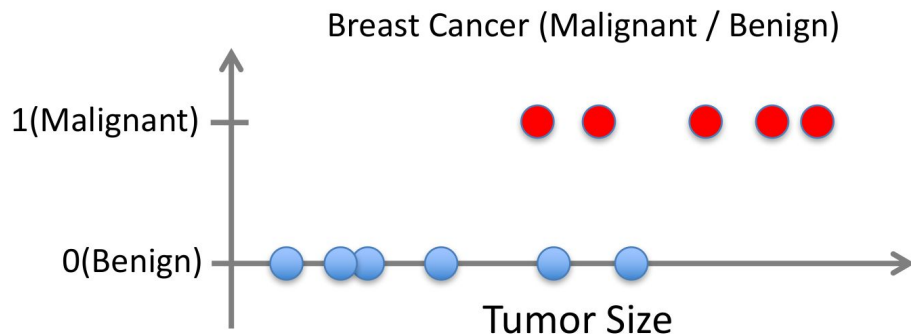


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Supervised Learning: Classification

- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
 - y is categorical == classification

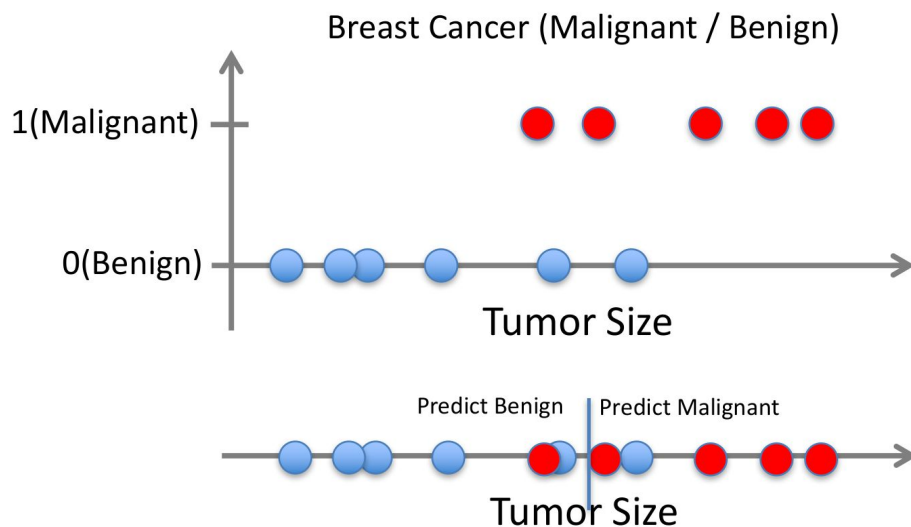
Binary Classification $y=\{-1,1\}$



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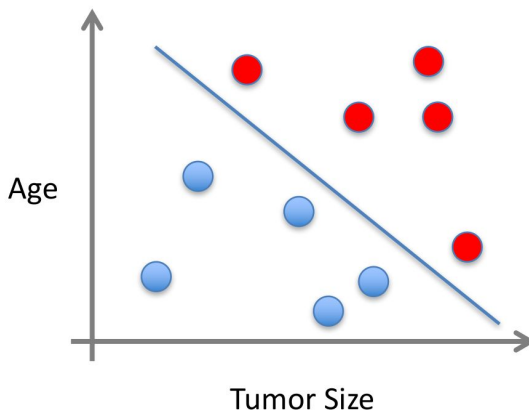
Binary Classification $y=\{-1,1\}$



Supervised Learning: Classification

- x can be multi-dimensional
 - Each dimension corresponds to an attribute

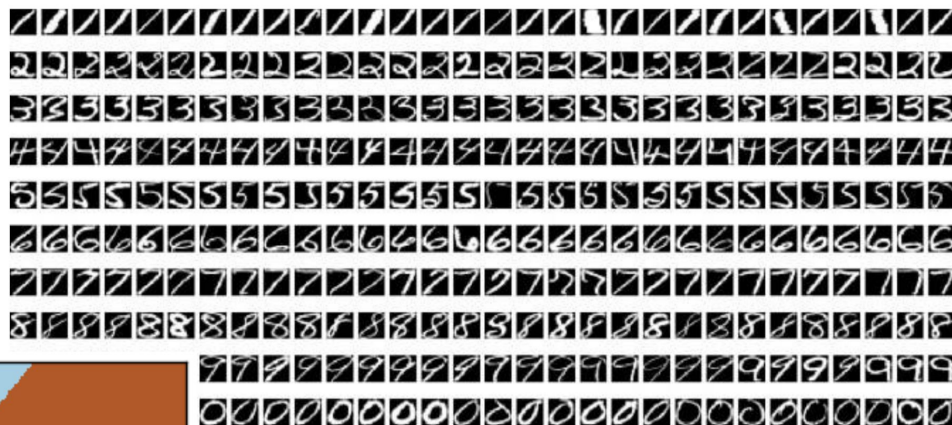
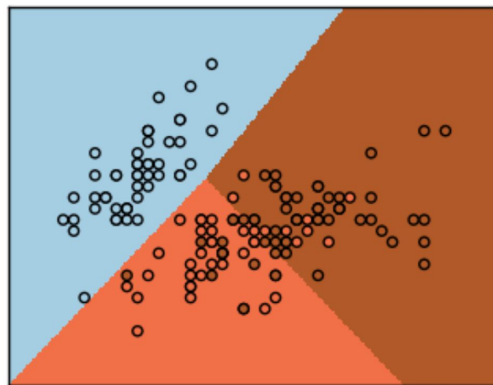
Binary Classification
 $y=\{-1,1\}$



- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape
- ...

Supervised Learning: Classification

Multiclass Classification
 $y=\{1,...,k\}$



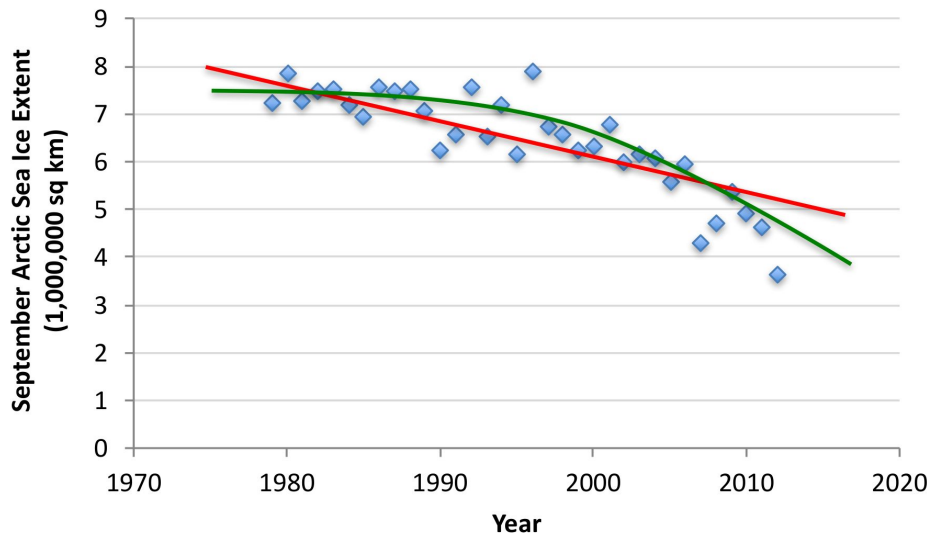
map image x to digit y

Supervised Learning: Regression

- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
 - y is real-valued == regression

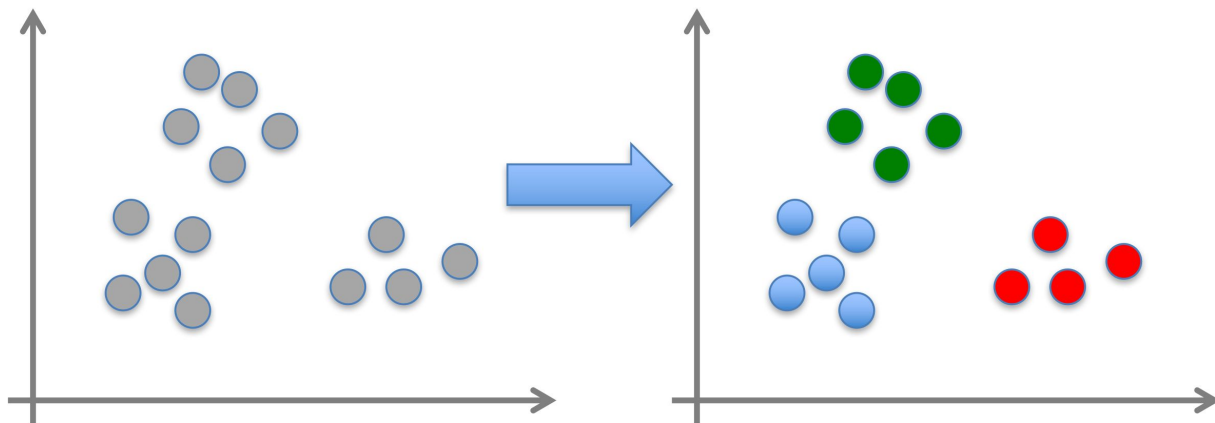
red: linear

green: non linear



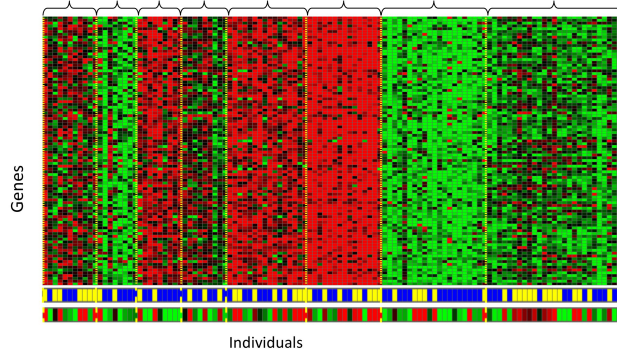
Unsupervised Learning

- Given $x_1, x_2, \dots, x_n$ (without labels)
- Output hidden structure behind the x 's
 - E.g., clustering

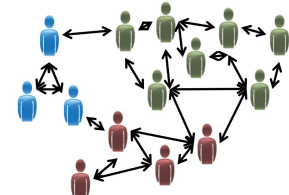


Unsupervised Learning

Genomics application: group individuals by genetic similarity



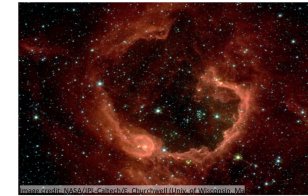
Organize computing clusters



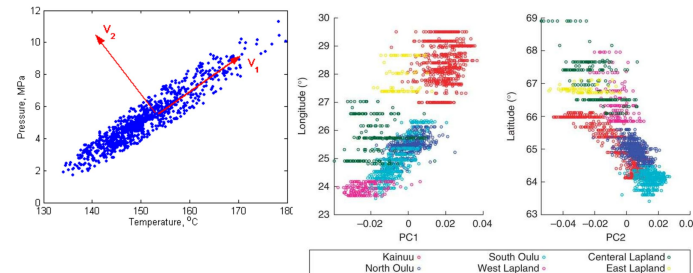
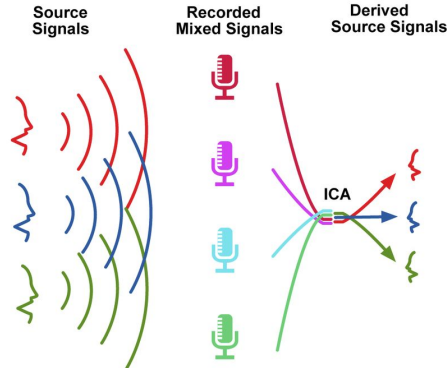
Social network analysis



Market segmentation



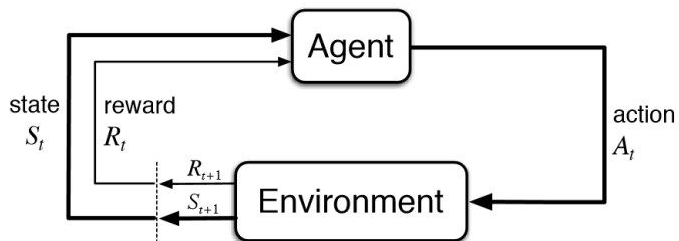
Astronomical data analysis



[Variance component model to account for sample structure in genome-wide association studies](#), Nature Genetics 2010

Many more Learnings...

- Reinforcement Learning



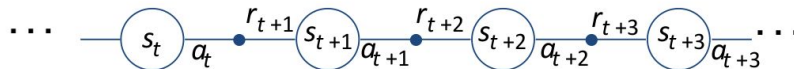
Agent and environment interact at discrete time steps : $t = 0, 1, 2, K$

Agent observes state at step t : $s_t \in \mathcal{S}$

produces action at step t : $a_t \in A(s_t)$

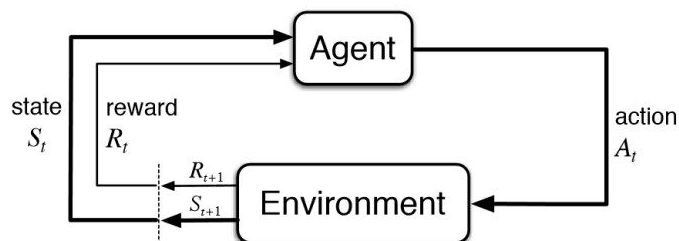
gets resulting reward : $r_{t+1} \in \mathcal{R}$

and resulting next state : s_{t+1}



Many more Learnings...

- Reinforcement Learning



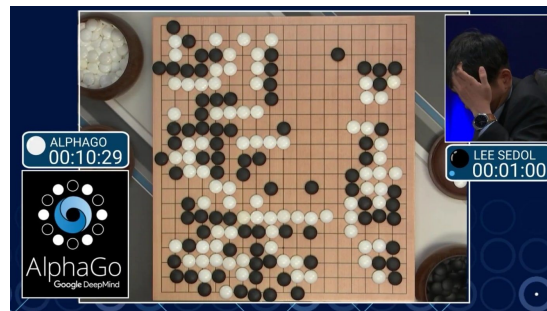
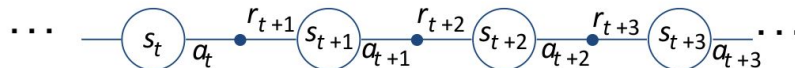
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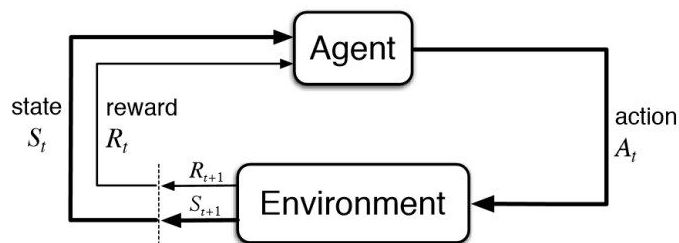
gets resulting reward : $r_{t+1} \in \mathcal{R}$

and resulting next state : s_{t+1}



Many more Learnings...

- Reinforcement Learning



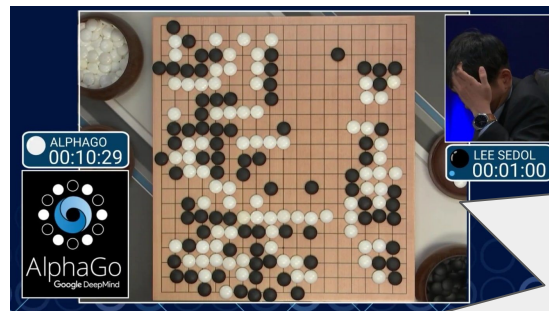
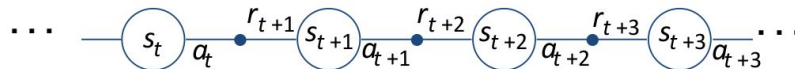
Agent and environment interact at discrete time steps : $t = 0, 1, 2, K$

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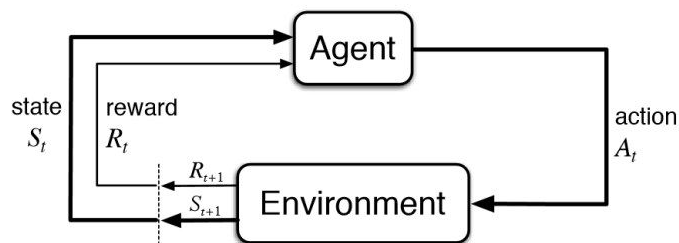
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Course
Robot
Learning
01HFNOV

Many more Learnings...

- Reinforcement Learning



- Active Learning
- Online Learning
- Multi-task Learning
- Metric Learning
- Ensemble Learning
- Self-Supervised Learning
- Discriminative vs Generative Learning
- [...]

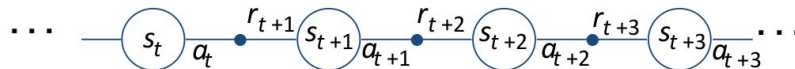
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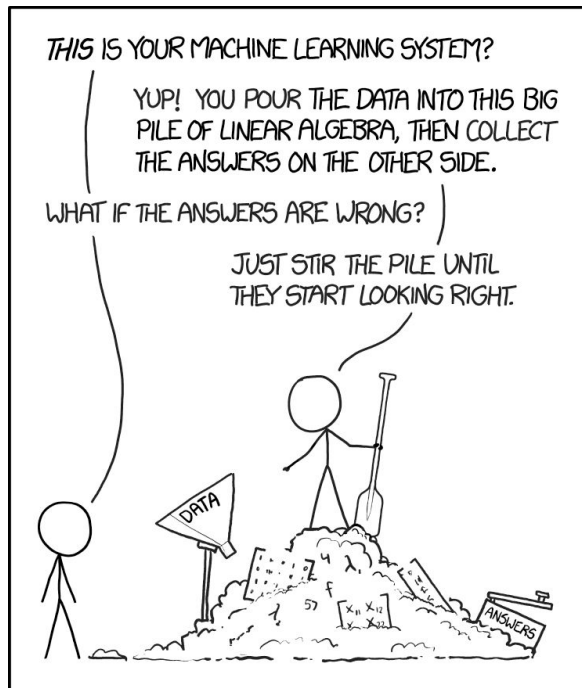


Who is a Machine Learner?



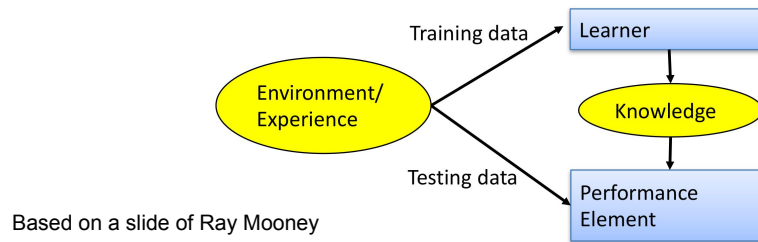
<https://xkcd.com/1838/>

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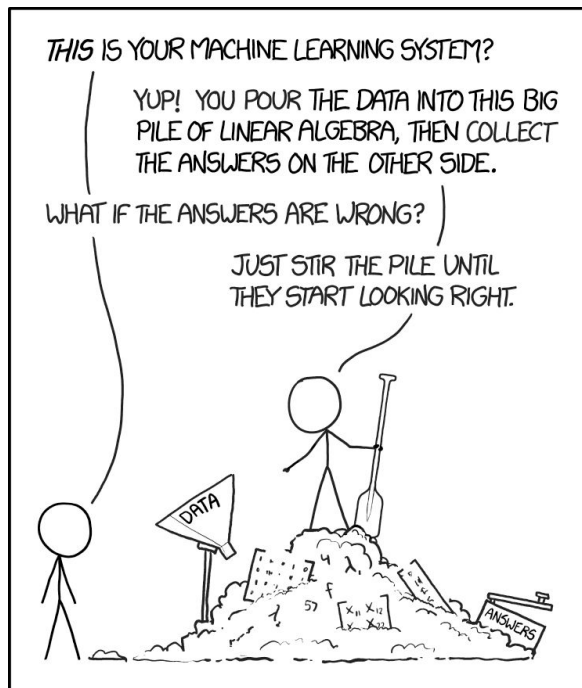
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- Framing a Learning Problem
- Designing a Learning System
 - choose the training experience
 - choose exactly what is to be learned (target function)
 - choose a learning algorithm to infer the target function from the experience



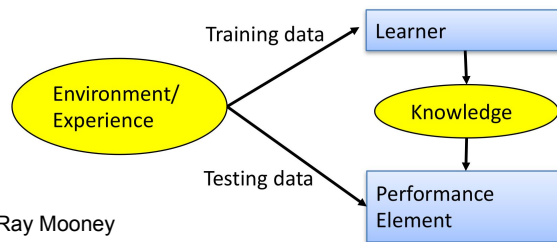
Based on a slide of Ray Mooney

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Based on a slide of Ray Mooney

Pseudo-Algorithm

1. Understand domain, prior knowledge, and goals
2. Data integration, selection, cleaning, pre-processing, etc.
3. Learn models
4. Interpret results
5. Consolidate and deploy discovered knowledge
6. Go back to 1

In just one minute online in 2025:

- 694M Spotify songs streamed
- 231M emails sent
- 6.3M Google searches conducted
- 3.47M YouTube videos watched
- 625M TikTok videos watched
- \$43.6M spent online during peak shopping times
- 174K apps downloaded
- 66K Instagram photos shared
- 2.1M active Facebook users



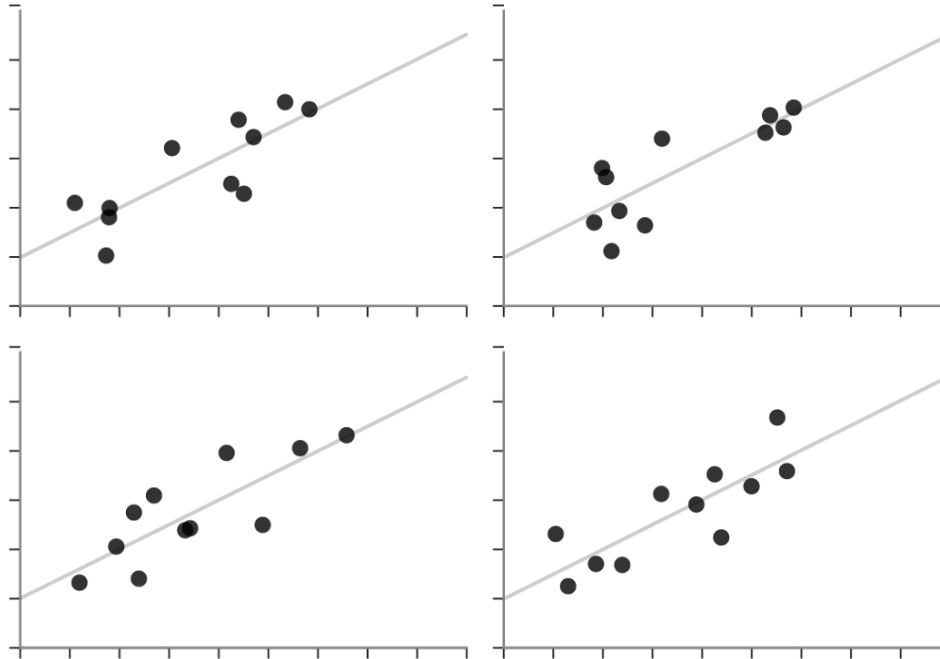
How to deal with data?

- **Observe the data**
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How to deal with data?

Unstructured Quartet

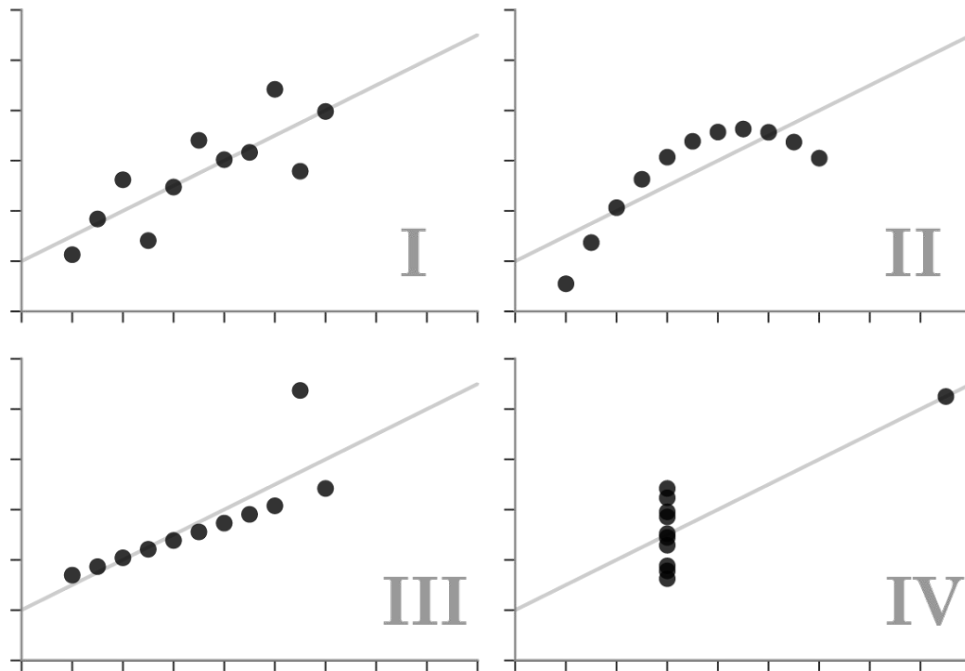
Each dataset here also has the same summary statistics. However, they are not *clearly different* or *visually distinct*.



How to deal with data?

Anscombe's Quartet

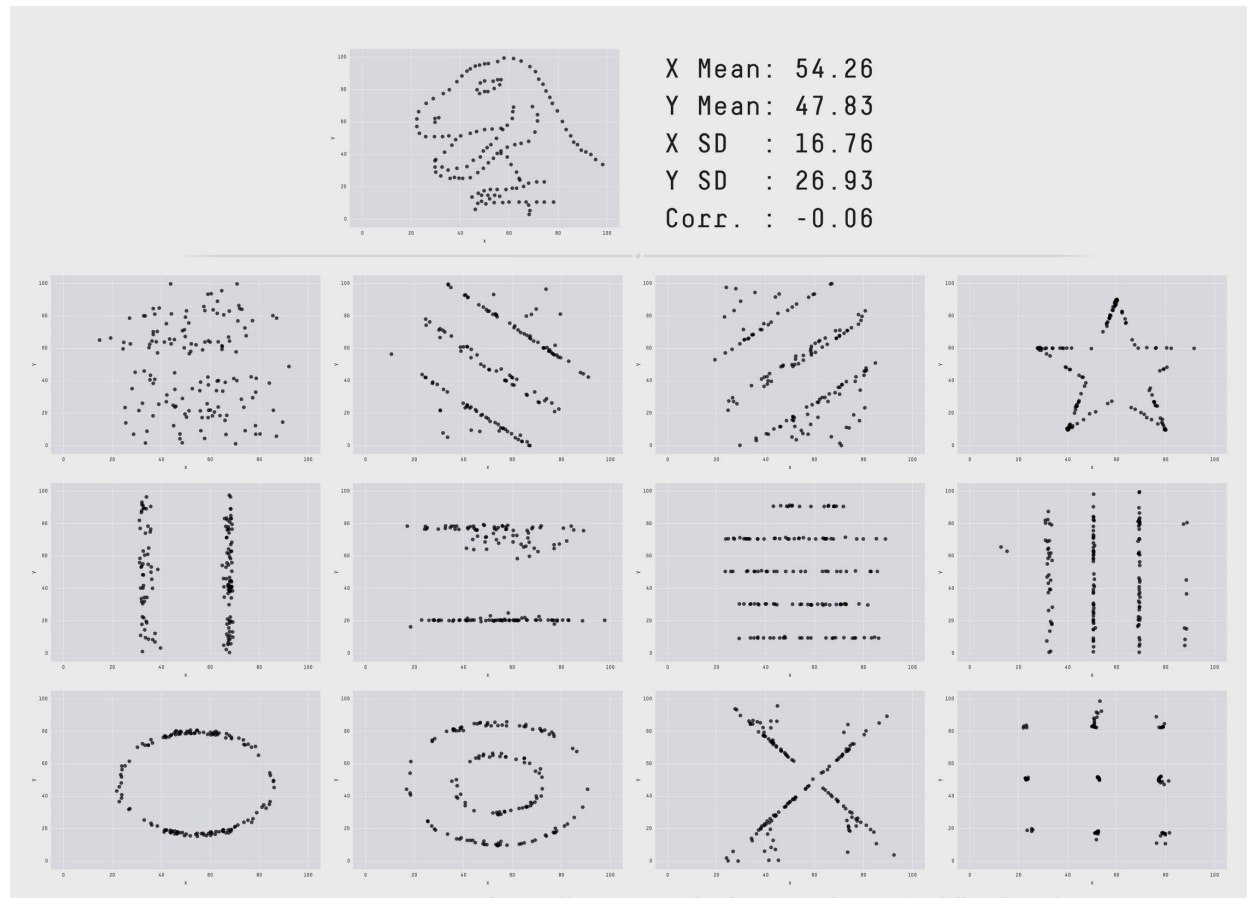
Each dataset has the same summary statistics (mean, standard deviation, correlation), and the datasets are *clearly different*, and *visually distinct*.



How to deal with data?

Datasaurus dataset

All these datasets
have the same
summary stats to 2
decimal places



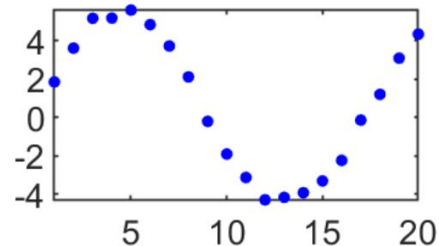
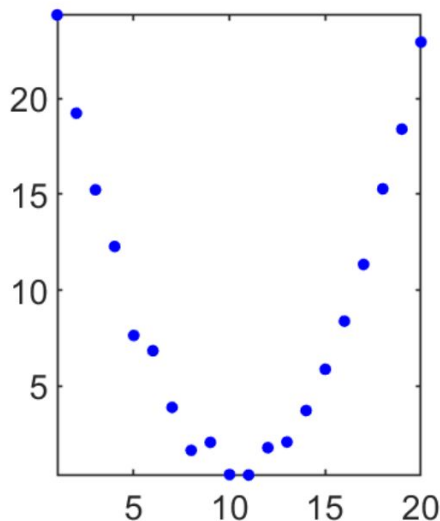
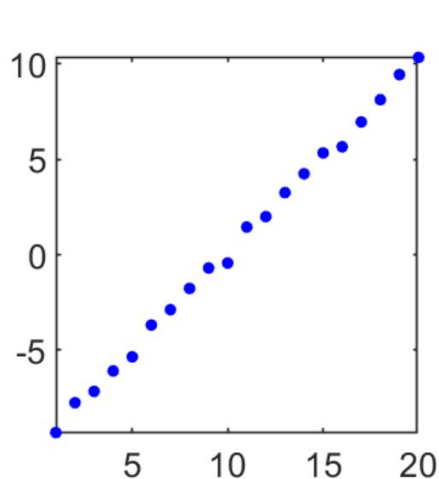
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Models for describing the data

Main idea: Look at the world, identify what knowledge we have about it, and use this knowledge to construct our model.

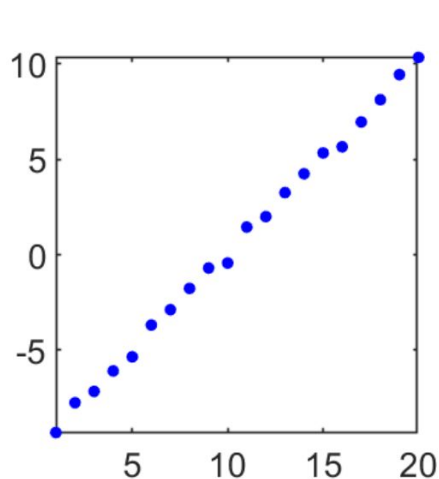
The model is a map from input (x) to output (y)



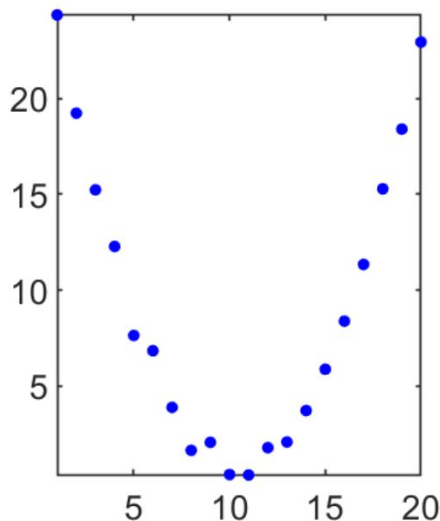
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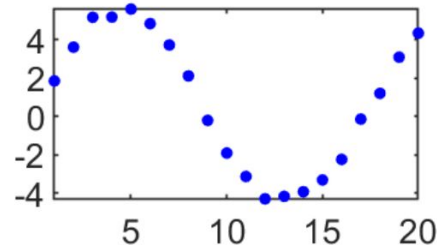
The model is a map from input (x) to output (y)



$$y = ax + b$$



$$y = ax^2 + bx + c$$

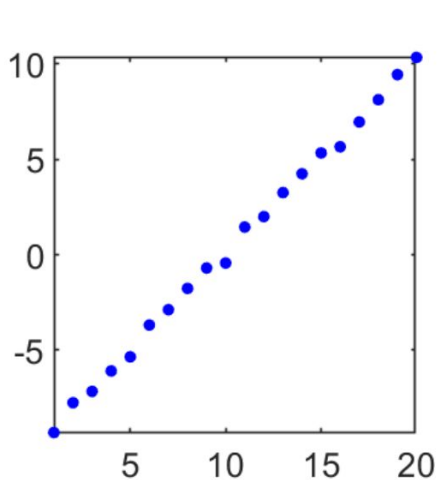


$$y = ax^3 + bx^2 + cx + d$$

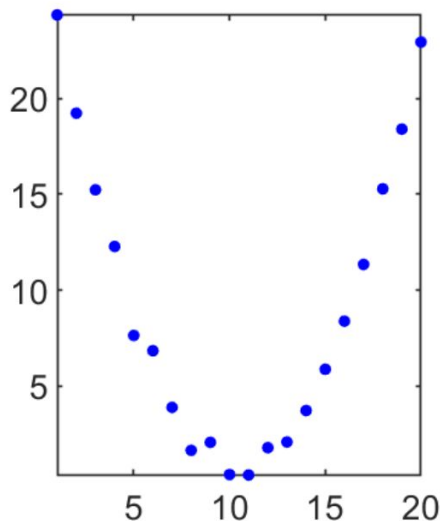
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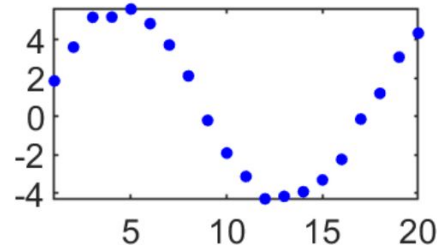
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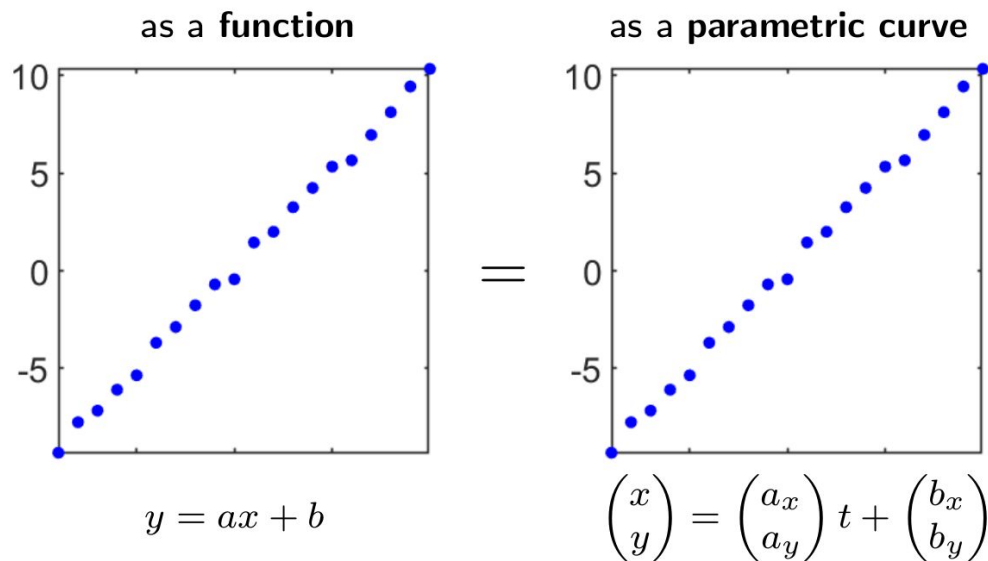
$$y = ax^2 + bx + c$$



$$y = a \sin(x) + bx + c$$

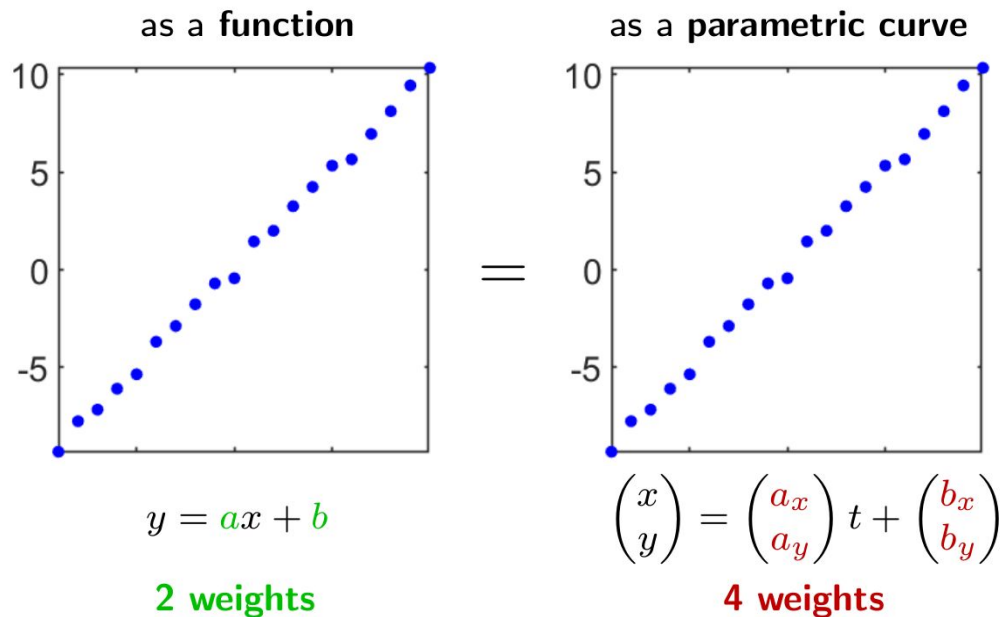
Choosing a Model Family

The same data can be described in different ways.
What is the “right” way?



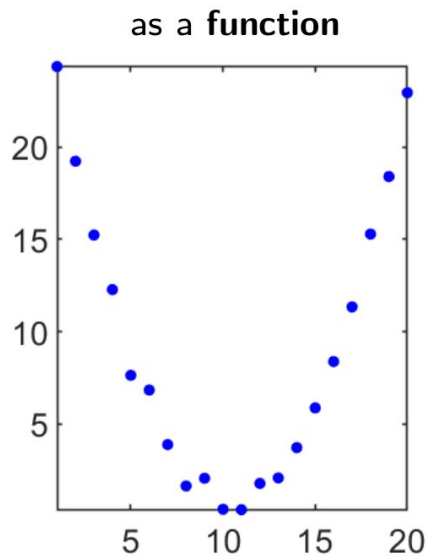
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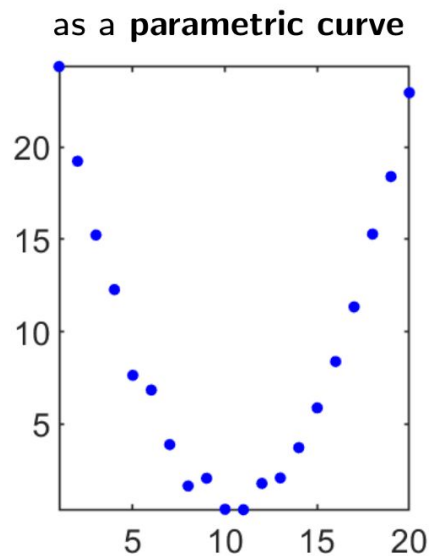
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$$y = ax^2 + bx + c$$

=

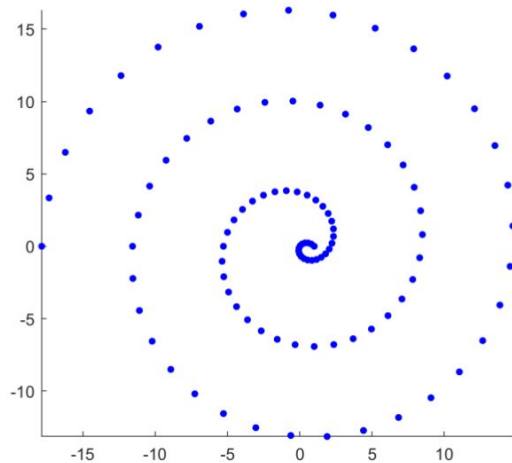


$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_x \\ a_y \end{pmatrix} t^2 + \begin{pmatrix} b_x \\ b_y \end{pmatrix} t + \begin{pmatrix} c_x \\ c_y \end{pmatrix}$$

Choosing a Model Family

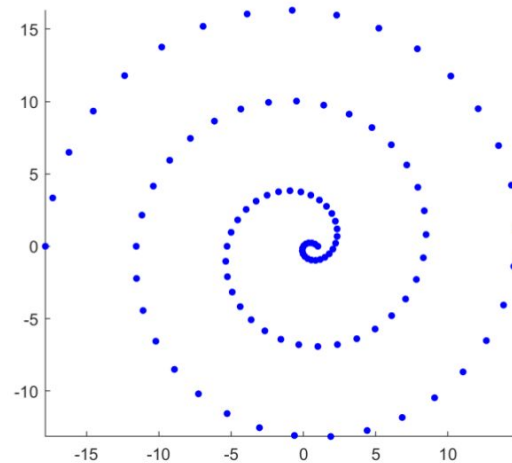
The same data can be described in different ways.
What is the “right” way?

as a **function**



y is not a function of x

as a **parametric curve**

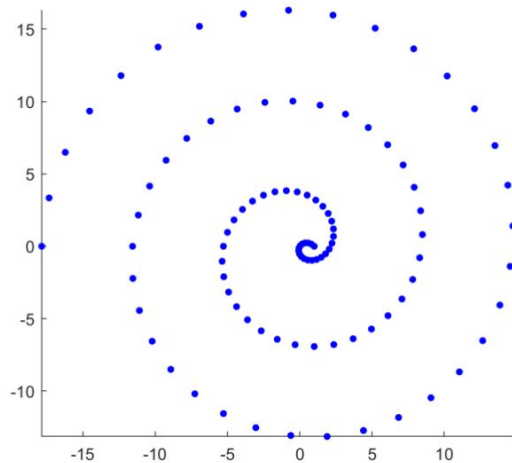


$$\begin{pmatrix} x \\ y \end{pmatrix} = a \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} (a - t)$$

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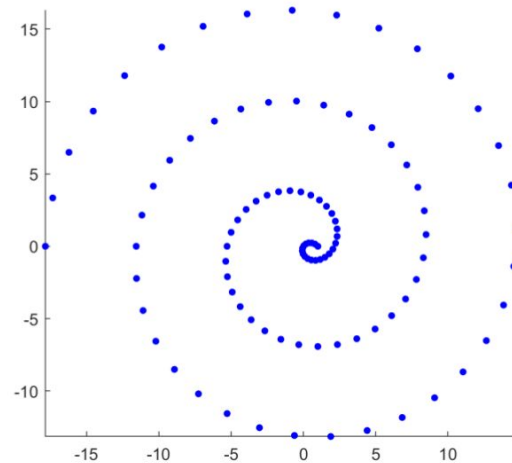
as a **function**



$$r = a\theta$$

(polar coordinates)

as a **parametric curve**

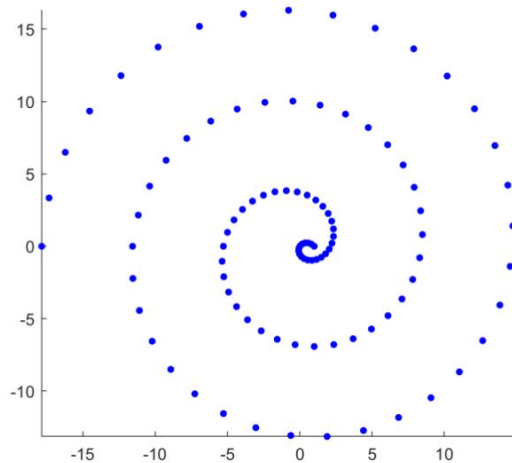


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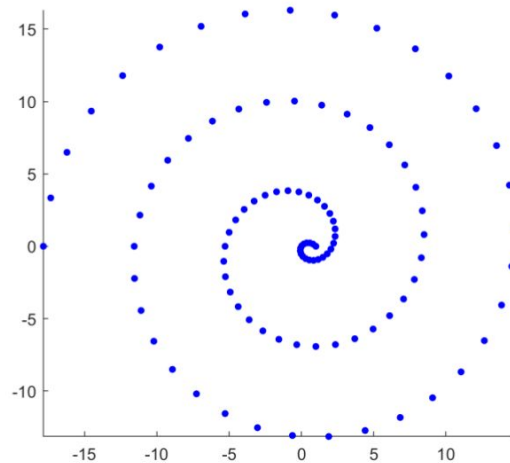
as a **function**



$r = a\theta$
(polar coordinates)

linear!

as a **parametric curve**

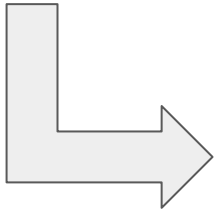


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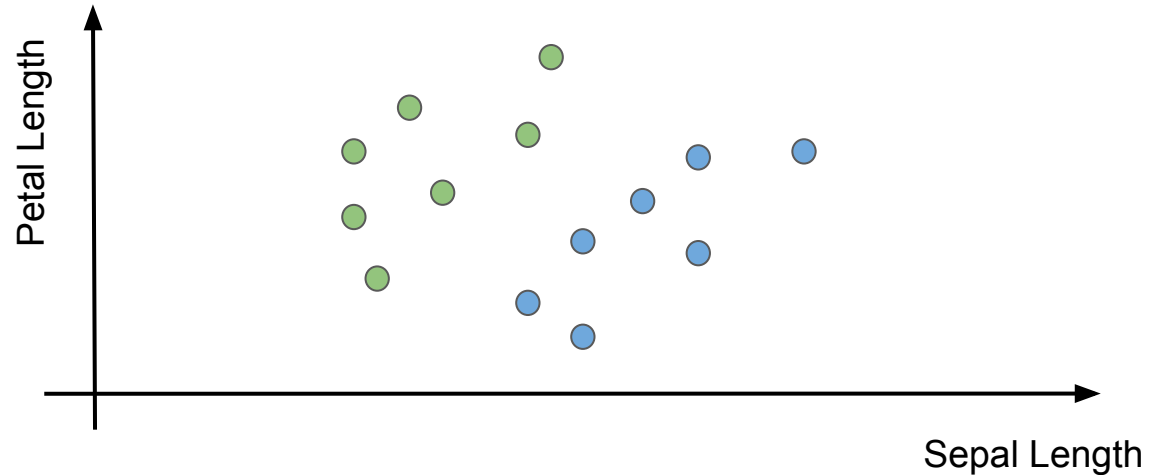
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Choosing a Representation



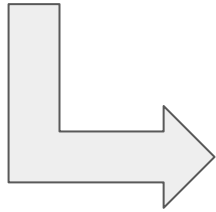
**Representation
Features**



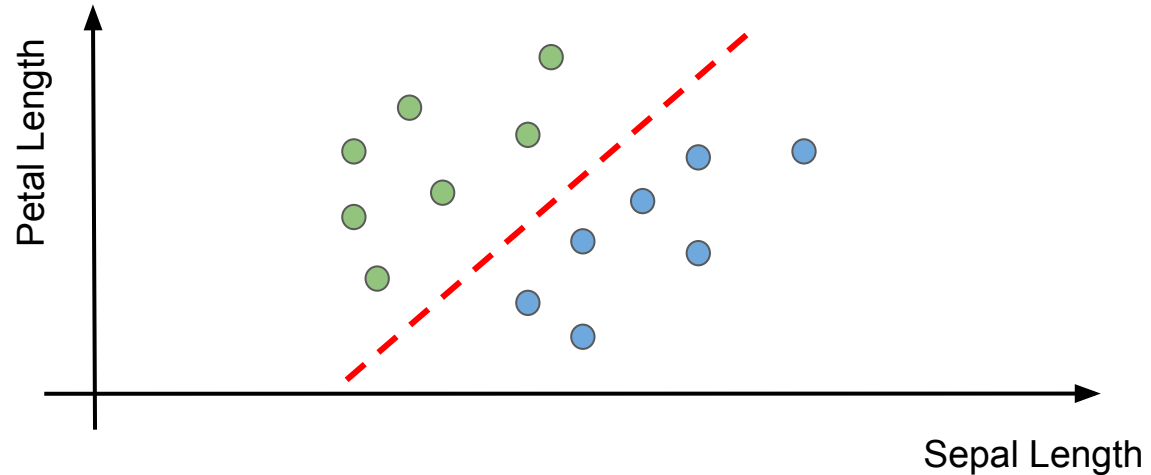
Choosing a Representation



Classification
Model



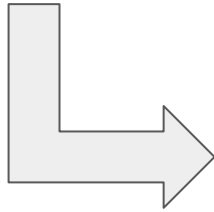
Representation
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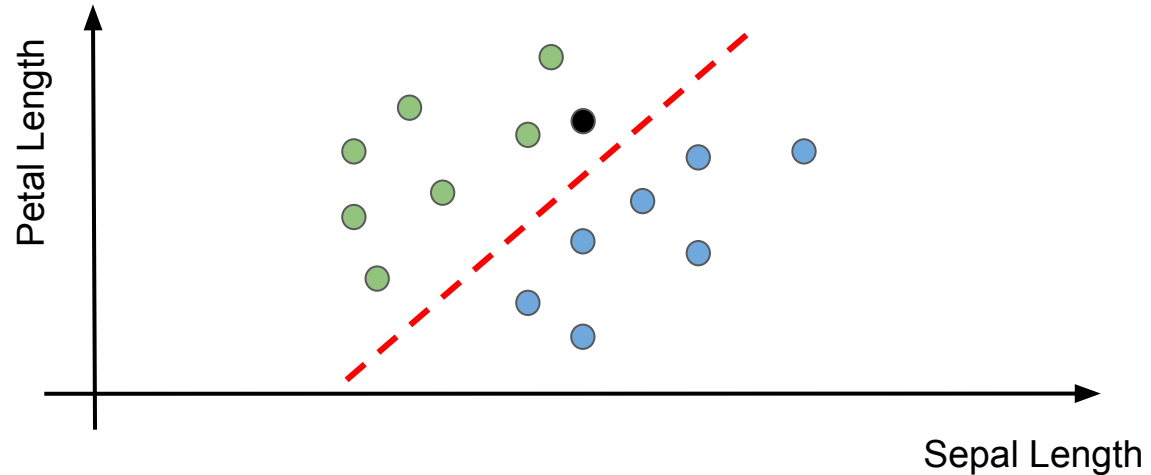
Choosing a Representation



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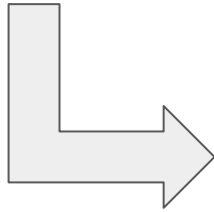
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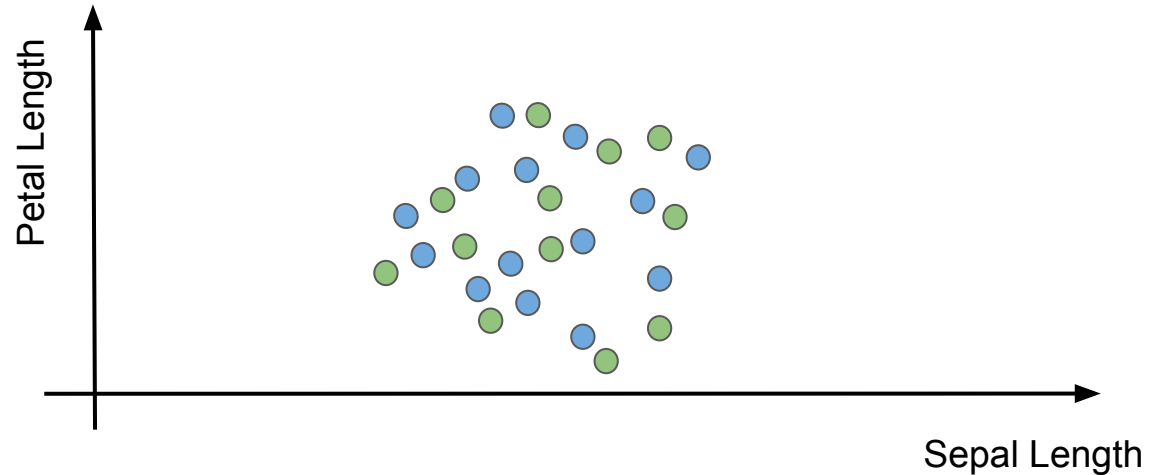
Choosing a Representation



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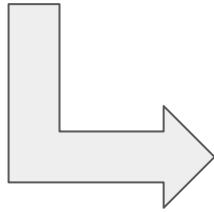
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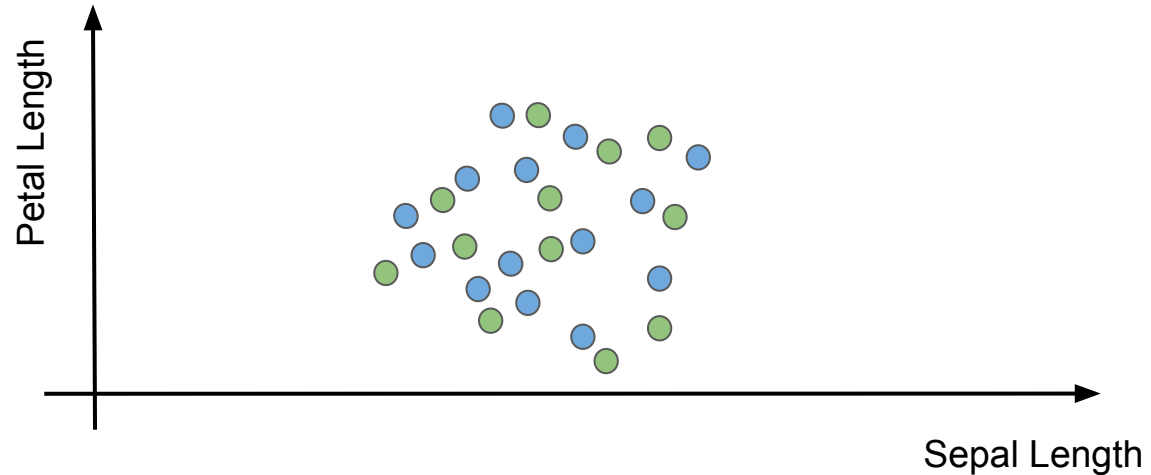
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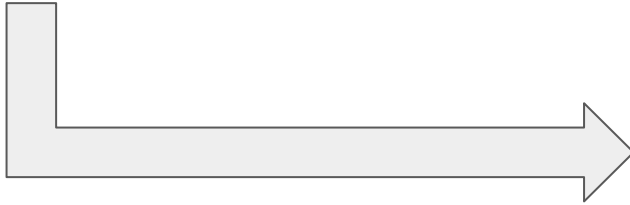
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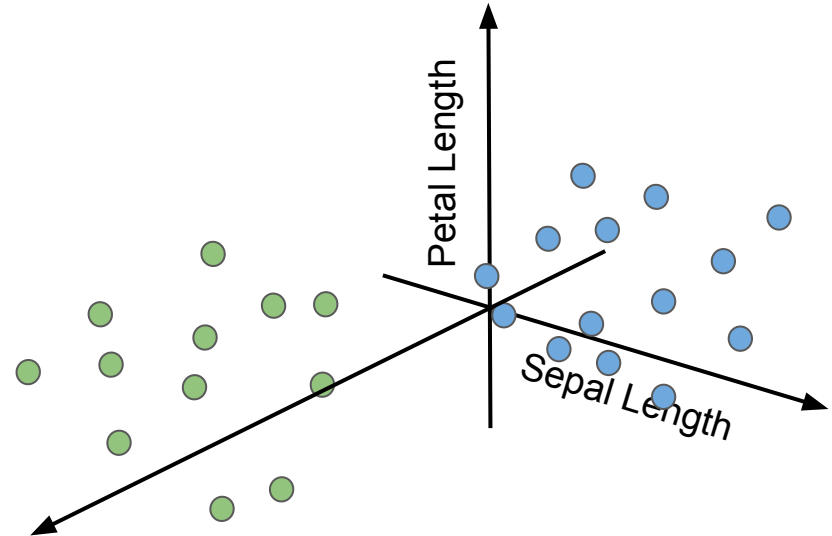


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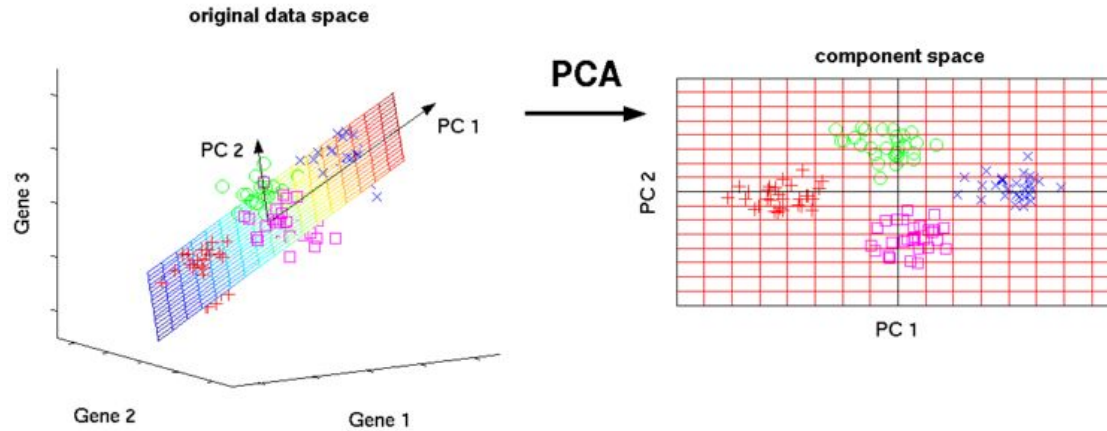


**Representation
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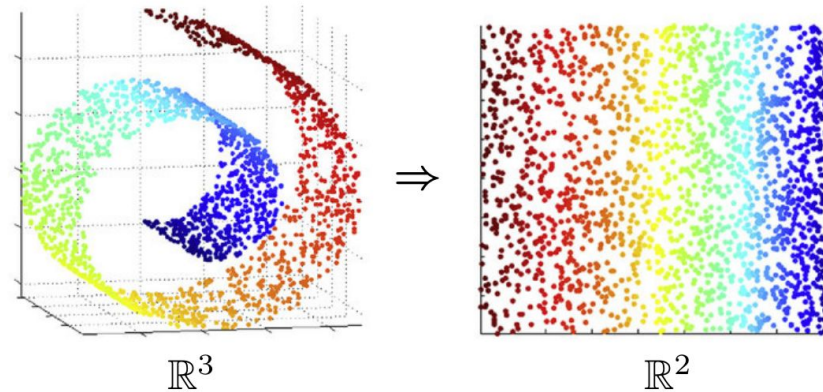
**Classification
Model**



Choosing a Representation



Linear and Non-linear
Dimensionality
Reduction



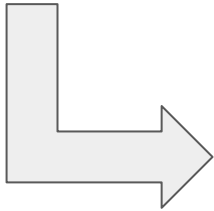
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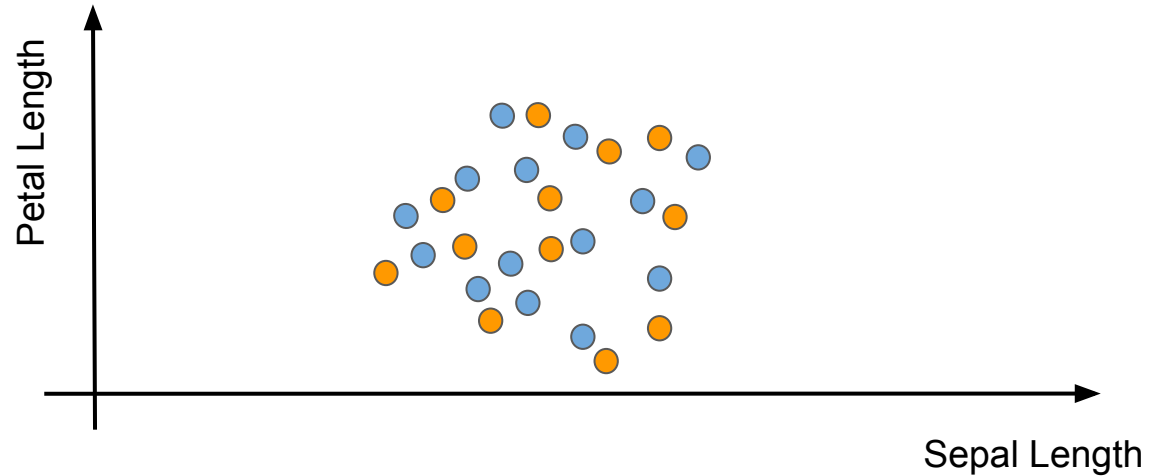
Deep Learning



**Classification
Model**



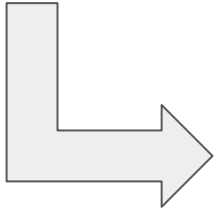
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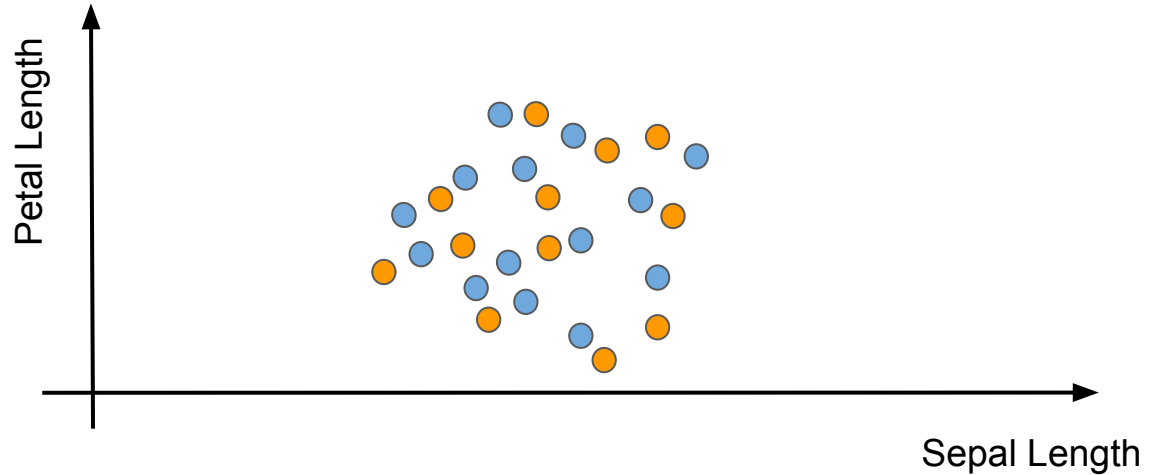
Deep Learning



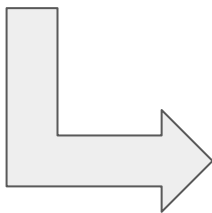
**Classification
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Deep Learning

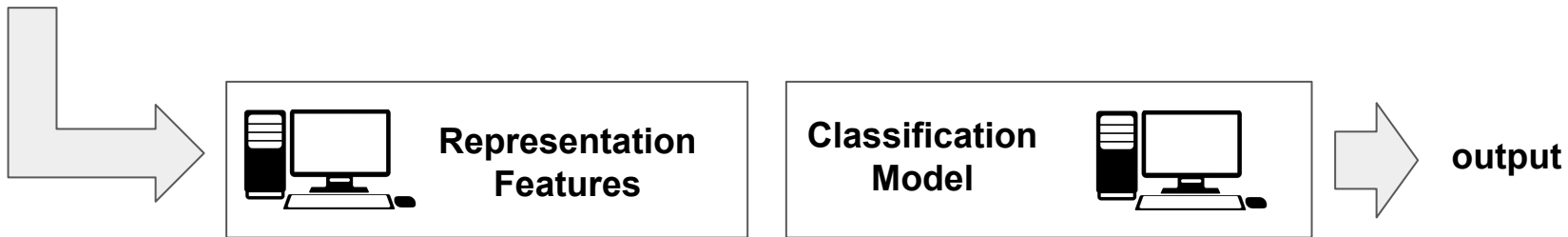


**Representation
Features**

**Classification
Model**



Deep Learning



Deep Learning

Features are Task-driven

Speaking about features only makes sense if we are given a task to solve!



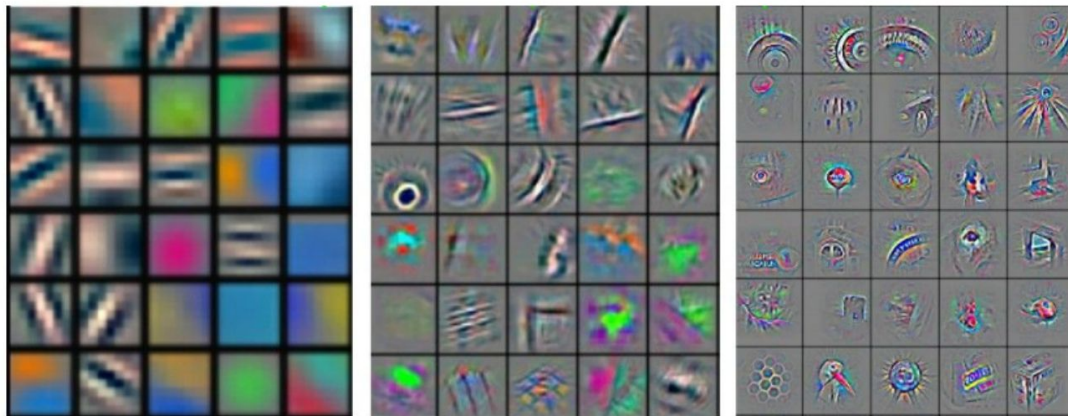
Is color important?

Rank, suit, and color are generic features, but specific problems determine what features are important for that task. What counts in spades, does not count in poker.

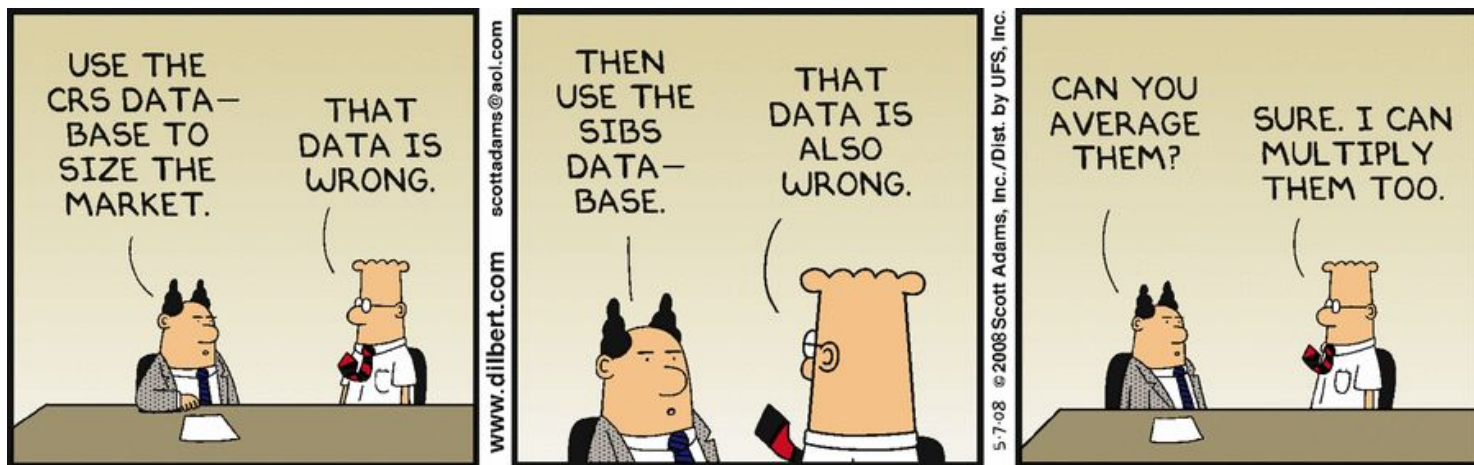
Deep Learning

Deep learning is a task-driven paradigm to extract patterns and latent features from given observations. However, features are not the focus of deep learning; rather, they are instrumental for the given task and drive the decision.

Example: Visual classification



Data Bias and Fairness



AI is objective only in the sense of learning what human teaches.

The data provided by human can be highly biased.

Data Bias and Fairness

	WHITE	AFRICAN AMERICAN
Labeled Higher Risk, But Didn't Re-Offend	23.5%	44.9%
Labeled Lower Risk, Yet Did Re-Offend	47.7%	28.0%

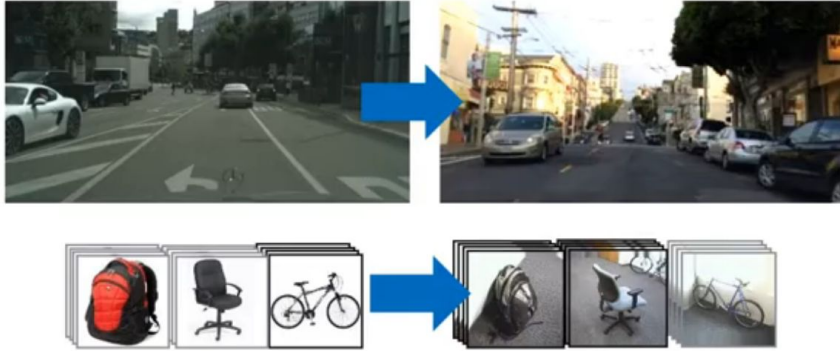
Overall, Northpointe's assessment tool correctly predicts recidivism 61 percent of the time. But blacks are almost twice as likely as whites to be labeled a higher risk but not actually re-offend. It makes the opposite mistake among whites: They are much more likely than blacks to be labeled lower risk but go on to commit other crimes. (Source: ProPublica analysis of data from Broward County, Fla.)

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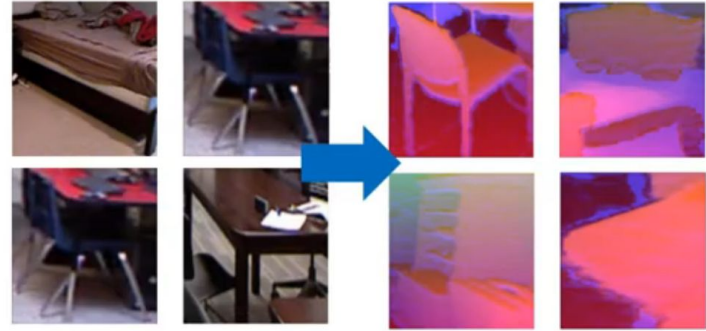
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Data Bias - Adaptation needed

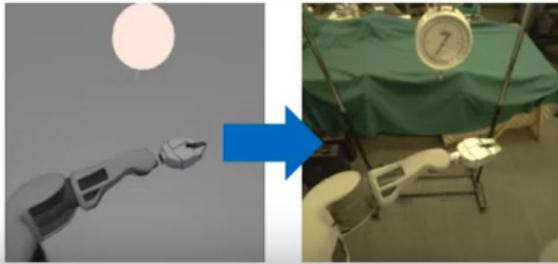
From dataset to dataset



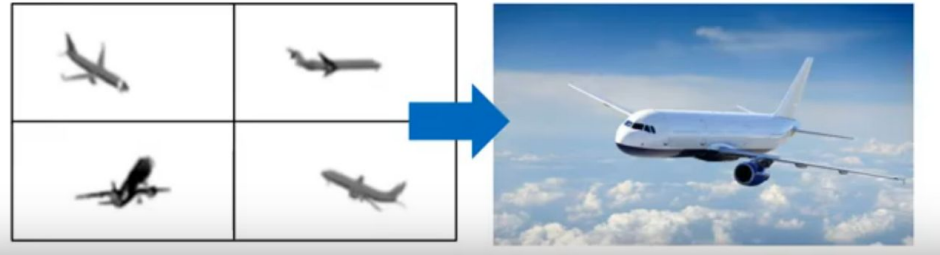
From RGB to depth



From simulated to real control

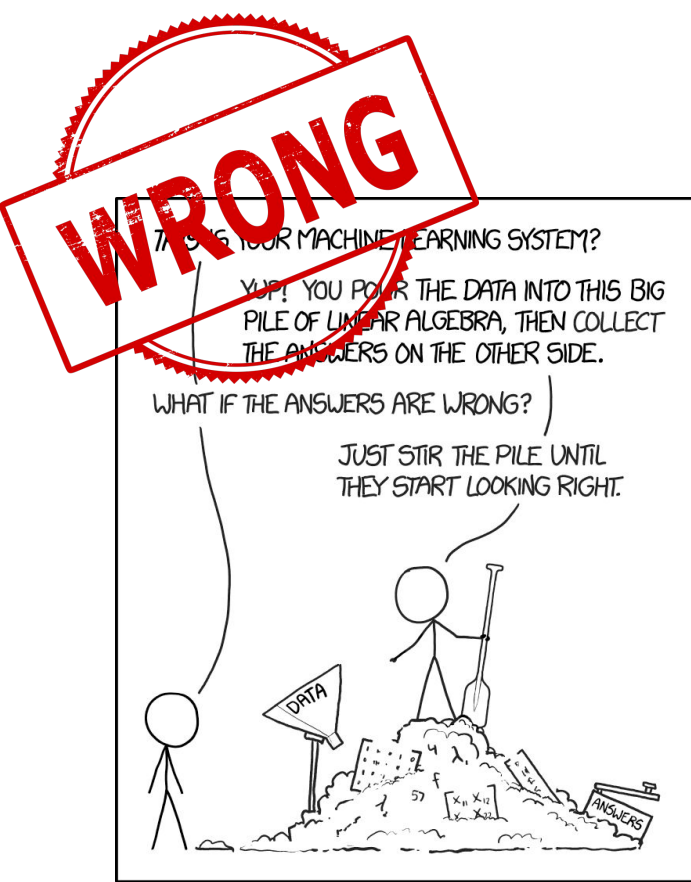


From CAD models to real images



Summary

- Course and Exam presentation
- What is
 - artificial intelligence
 - machine learning
 - deep learning
- Learning: build a model
- How to deal with data
 - observe samples
 - feature representation
 - dataset bias and fairness



<https://xkcd.com/1838/>

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Probabilistic reasoning